

Generalized Biomedicine Discovery

Luyao Tang · Yingkai Yang · Hanqi Chen · Jiewei Zheng · Chaoqi Chen · Cheng Chen†

The University of Hong Kong · Shenzhen University · Xiamen University

Code: github.com/lytang63/generalized-biomedicine-discovery

Contact: lytang1999@gmail.com

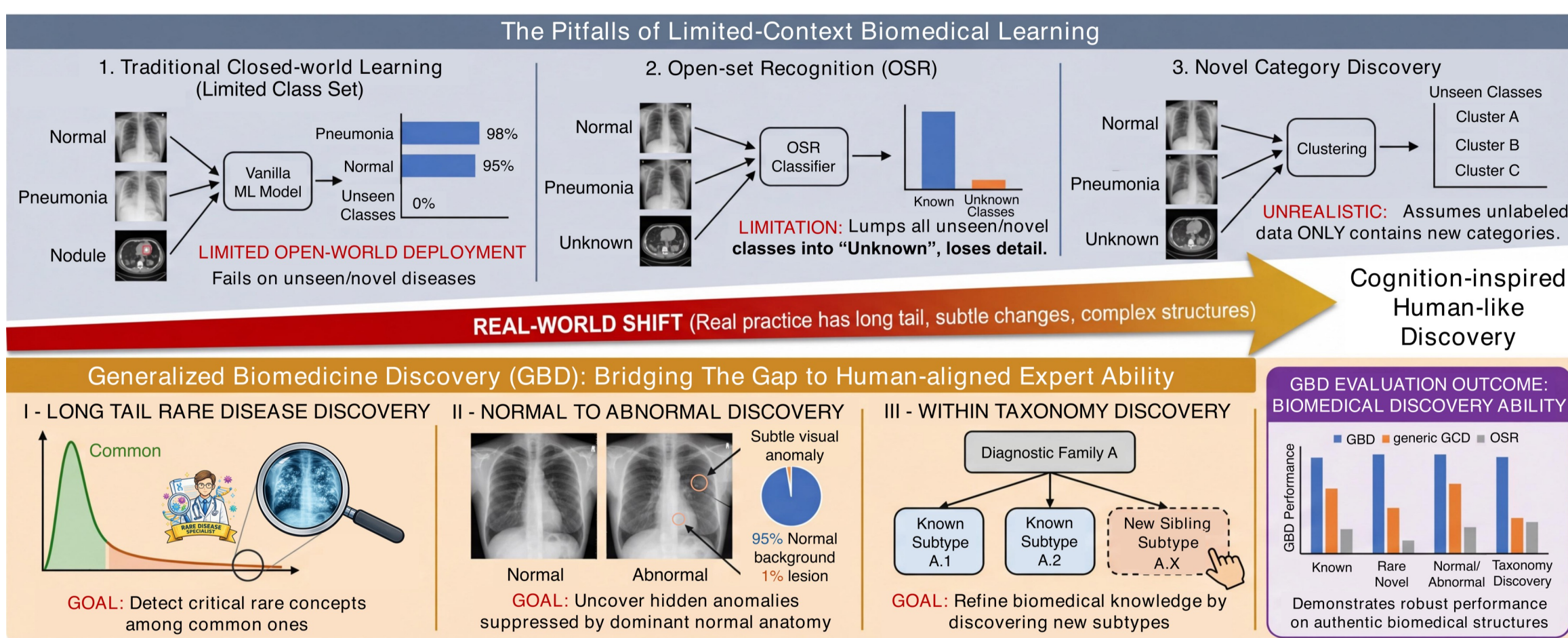
1 · Why GBD Is Harder Than Standard Category Discovery

Clinical gap:

Clinical archives are long-tailed, dominated by normal anatomy, and organized by diagnostic taxonomies - unlike balanced, flat open-world benchmarks.

GBD reframing:

Discover clinically meaningful novelty while labeled known classes and unlabeled known/novel cases coexist.



THREE REAL-WORLD SHIFTS

- I. Long-tail:** rare diseases are represented by common head classes.
- II. Normal-to-abnormal:** small lesions are overwhelmed by normal anatomy.
- III. Within-taxonomy:** novel sibling subtypes resemble known relatives.

2 · Task, Data, and Evaluation

$$\mathbf{Y}_I = \mathbf{C}_{\text{known}}; \mathbf{Y}_u = \mathbf{C}_{\text{known}} \cup \mathbf{C}_{\text{novel}}$$

- Train on labeled $\mathbf{D}_I$ and mixed unlabeled $\mathbf{D}_u$; $\mathbf{D}_u$ targets are evaluation-only.
- One Hungarian assignment method evaluates All / Known / Novel accuracy.
- Gastrovision-endoscopy · HistoSet-pathology · MLL23-microscopy · Derm12345-dermatology.

$$\text{ACC} = \max_{p \in \mathcal{P}(\mathcal{Y}_u)} \frac{1}{|\mathcal{D}_u|} \sum_{i=1}^{|\mathcal{D}_u|} \mathbb{1}\{y_i^u = p(\hat{y}_i)\}.$$

ACC asks whether predicted clusters recover the mixed known+novel label space after optimal label alignment; no unlabeled ground truth enters training.

3 · SCAN Logic and Variables

1 PREDICTIVE SUPPRESSION

Remove from each patch token $\mathbf{x}_i$ the component explained by dominant anchor $\mathbf{u}$; residual $\mathbf{r}_i$ retains under-explained evidence.

$$\mathbf{r}_i = \mathbf{x}_i - (\mathbf{x}_i^\top \text{sg}(\mathbf{u})) \text{sg}(\mathbf{u}), \quad \mathbf{R} = [\mathbf{r}_1, \dots, \mathbf{r}_N]^\top.$$

2 STABLE ACCOMMODATION

Project novelty away from $\mathbf{u}$, then use gate γ to inject complementary evidence; $\gamma \approx 0$ leaves class token $\mathbf{c}$ unchanged.

$$\tilde{\mathbf{c}} = \mathbf{c} + \gamma \cdot \text{LN}(\Delta_\perp + \mathbf{x}_{\text{pool}\perp})$$

VARIABLE MAP

$\mathbf{x}_i$: patch token · $\mathbf{u}$: dominant anchor · $\mathbf{r}_i$: residual · $\mathbf{w}_i$: salience weight
 $\Delta_\perp$: orthogonal novelty · $\mathbf{x}_{\text{pool}\perp}$: pooled complementary evidence · γ : surprise gate · $\tilde{\mathbf{c}}$: updated token

LOGIC: predictable evidence down -> residual surprise up -> complementary evidence added without erasing known structure.

4 · SCAN: From Dominant Patterns to Complementary Evidence

One feature-space layer, three linked operations, unchanged host GCD objective.

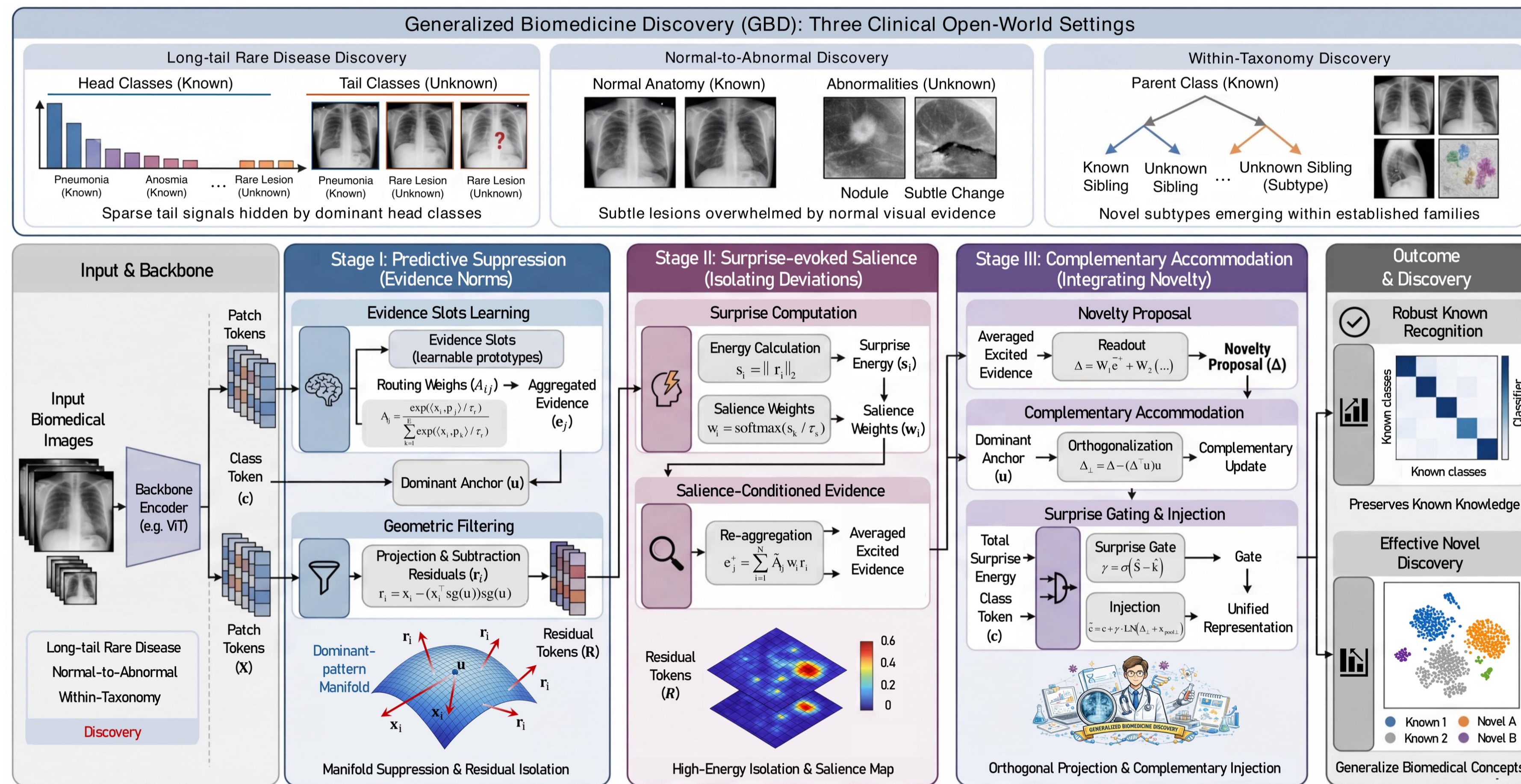


Figure 2 - suppress what the model already predicts -> score residual surprise -> inject only the orthogonal complement.

- Route N patch tokens into evidence slots and aggregate sample-specific anchor $\mathbf{u}$.
- Remove anchor-aligned components to isolate residual token matrix $\mathbf{R}$.
- Convert residual energy s_i into salience w_i and re-aggregate excited evidence.
- Orthogonalize novelty proposal Δ , then gate its injection with γ to preserve known knowledge.

WHY IT WORKS: residualization weakens dominant-feature suppression; orthogonal gating preserves an identity path.
DEPLOYMENT: SCAN operates in feature space while the host backbone and GCD objective remain unchanged.

5 · Results I-II: Gains Across Modalities and Backbones

WHY SCAN FITS SETTINGS I-II

Both settings hide clinically decisive evidence behind a dominant known pattern;
 SCAN isolates what that pattern fails to explain.

Table 1: Comparison on **Setting I** (Long-tail Rare Disease Discovery) across endoscopy, pathology, microscopy, and dermatology images. Results are accuracy on All, Old, and New subsets, with the best results highlighted in bold.

Method	Gastrovision			HistoSet			MLL23			Derm12345			Average		
	All	Old	New	All	Old	New	All	Old	New	All	Old	New	All	Old	New
SEAL†	56.8	51.2	60.7	81.2	87.1	77.9	65.4	67.6	54.1	56.5	61.0	10.6	65.0	66.7	50.8
+ SCAN	71.2	83.4	53.0	76.3	99.9	69.7	80.7	89.3	58.6	35.6	35.6	66.0	77.1	54.2	54.2
SimGCD	37.8	40.6	31.8	74.4	83.8	71.8	56.9	52.9	67.0	25.4	27.9	13.1	48.6	51.3	45.9
+ SCAN	39.7	43.3	31.9	80.0	82.1	79.4	64.3	63.1	67.2	25.9	28.2	14.2	52.5	54.2	48.1
Δ	1.9	2.7	0.1	5.6	-1.7	7.6	7.4	10.2	0.2	0.5	0.4	1.1	3.8	2.9	2.2
LegoGCD	39.8	48.9	26.1	72.5	76.0	71.6	70.8	63.4	89.7	26.3	28.5	15.7	52.3	54.2	50.8
+ SCAN	41.3	49.4	29.2	72.5	77.5	71.2	73.2	67.2	88.8	26.7	29.0	14.8	53.4	55.8	51.0
Δ	1.5	0.5	3.1	0.0	1.5	-0.4	2.4	3.8	-0.9	0.3	0.6	-0.9	1.1	1.6	0.2
Avg. Δ	8.2	11.5	2.9	1.8	0.7	2.1	8.1	11.6	-1.2	2.5	2.5	2.7	5.1	6.6	1.6

Table 2: Comparison on **Setting II** (Normal-to-Abnormal Discovery) across endoscopy, pathology, microscopy, and dermatology images. Results are accuracy on All, Old, and New subsets, with the best results highlighted in bold.

Method	Gastrovision			HistoSet			MLL23			Derm12345			Average		
	All	Old	New	All	Old	New	All	Old	New	All	Old	New	All	Old	New
SEAL†	62.6	64.2	61.0	73.5	98.7	48.2	69.2	70.9	60.1	59.0	61.7	5.5	66.1	73.8	43.7
+ SCAN	70.1	86.0	46.3	68.6	97.2	60.7	87.1	89.1	81.7	33.6	33.3	36.4	76.4	56.3	56.3
Δ	19.8	34.1	4.6	4.4	-0.5	13.1	28.6	31.1	21.9	4.3	5.1	-4.3	14.2	17.5	8.8
SimGCD	39.7	42.9	25.6	61.1	94.1	44.5	57.9	55.9	63.4	26.6	27.2	19.9	46.3	55.0	38.3
+ SCAN	40.1	44.0	22.6	62.0	93.6	46.1	58.3	57.5	60.6	26.4	26.8	21.6	46.7	55.5	37.7
Δ	0.4	1.1	-3.0	0.9	-0.6	1.6	0.5	1.7	-2.8	-0.2	-0.4	1.7	0.4	0.5	-0.6
LegoGCD	49.3	53.7	24.9	62.0	97.5	44.2	68.3	72.1	58.6	28.5	30.8	5.5	52.1	63.5	33.3
+ SCAN	50.5	55.5	23.1	63.9	97.4	47.1	69.4	73.7	57.7	29.0	29.7	21.6	53.2	64.1	37.4
Δ	1.3	1.8	-1.8	1.9	-0.1	2.9	0.9	1.6	-0.9	0.4	-1.1	16.1	1.1	0.6	4.1
Avg. Δ	7.1	12.3	-0.1	2.4	-0.4	5.9	10.0	11.5	6.1	1.5	1.2	4.5	5.2	6.2	4.0

Selex + SCAN (Ours):
 Rare tail: All 55.5 → 66.0 (+10.5)
 Subtle abnormality: All +14.2 · Novel +8.8

SETTING I (Long-tail) — Common head-class evidence can dominate clustering. Suppression removes that direction so rare morphology survives in the residual; surprise weighting then surfaces tail cues. Gastrovision Old +31.4.

SETTING II (Normal-to-abnormal) — Normal anatomy is the dominant anchor. Residual salience emphasizes small lesion deviations, while orthogonal gated injection preserves normal recognition. MLL23 All +28.6 / Novel +21.9.

READOUT — Gains are broad rather than universal; flat or negative host/subset cells mark dependence on what the backbone still encodes.

6 · Result III: Sibling-Subtype Discovery

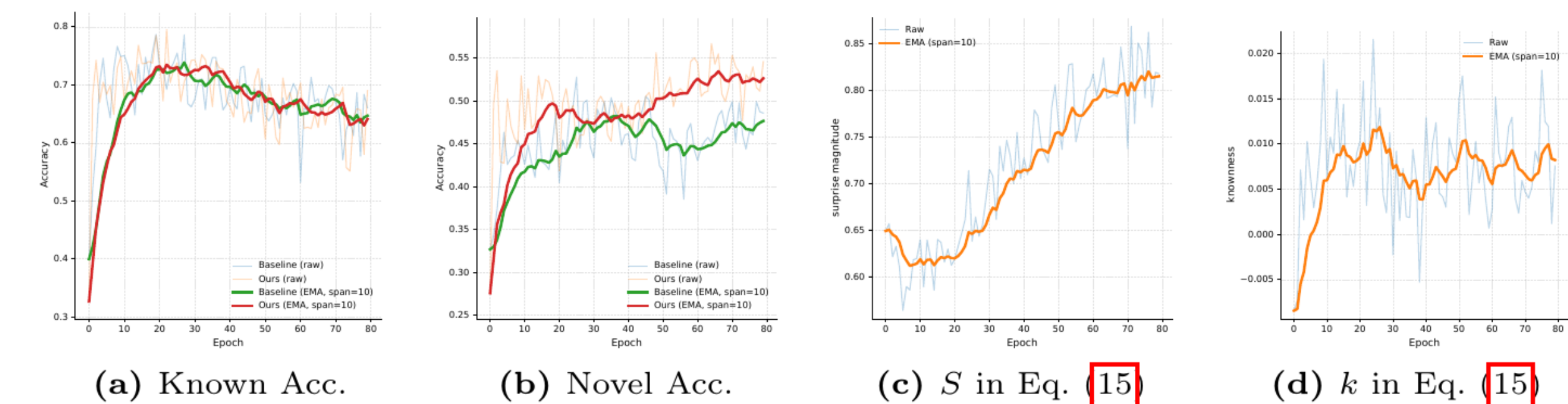
SCIENTIFIC QUESTION
 When parent anatomy is shared,
 which residual morphology separates an unseen sibling?

Method	Gastrovision			HistoSet			MLL23			Derm12345			Average		
	All	Old	New	All	Old	New	All	Old	New	All	Old	New	All	Old	New
SEAL†	54.9	51.3	64.3	92.9	95.7	89.3	70.6	83.3	47.2	27.7	62.4	16.4	61.5	73.2	54.3
Selex	50.1	45.7	53.8	77.5	99.4	71.4	62.5	69.9	55.6	24.8	65.1	18.2	53.7	70.0	49.8
+ SCAN	66.2	79.6	46.3	88.5	96.4	83.2	81.4	89.0	61.9	27.8	69.9	20.9	66.0	83.7	53.1
Δ	16.1	33.8	-7.5	11.0	-3.0	11.7	18.9	19.0	6.3	3.0	4.8	2.7	12.3	13.7	3.3
SimGCD	42.9	43.0	42.5	89.4	91.7	87.9	53.2	57.4	49.4	19.3	40.8	15.9	51.2	58.2	48.9
+ SCAN	42.0	42.4	39.4	89.1	90.5	88.2	54.0	58.4	49.9	17.8	36.8	14.7	50.7	57.0	48.1
Δ	-0.9	-0.6	-3.1	-0.3	-1.2	0.2	0.8	1.1	0.5	-1.5	-4.0	-1.2	-0.5	-1.2	-0.8
LegoGCD	40.2	39.6	40.6	75.8	84.4	70.0	54.7	61.8	48.2	19.4	41.0	16.0	47.5	56.7	43.7
+ SCAN	40.5	40.4	40.6	80.5	86.2	76.6	60.3	68.7	52.6	20.8	42.4	17.2	50.5	59.4	46.8
Δ	0.4	0.8	0.0	4.7	1.8	6.6	5.6	6.9	4.4	1.3	1.5	1.3	3.0	2.8	3.1
Avg. Δ	5.2	11.3	-3.6	5.1	-0.8	6.2	8.4	9.0	3.7	0.9	0.8	0.9	4.9	5.1	1.8

DESIGN BENEFIT — Subtracting the parent-aligned direction exposes subtype-specific evidence; orthogonal accommodation preserves family and known-sibling structure while opening cluster separation.

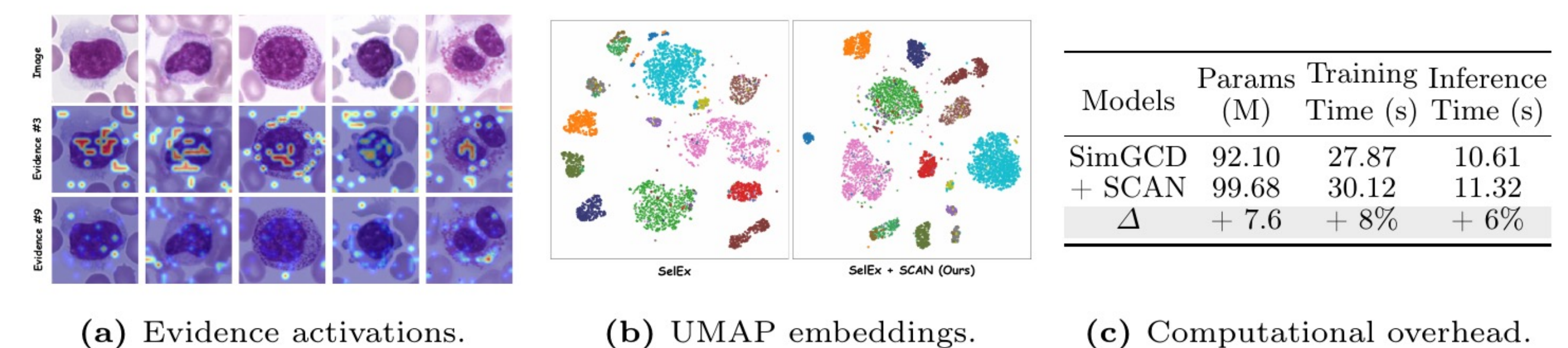
EVIDENCE (Within-taxonomy) — Selex Avg. All 53.7 → 66.0 (+12.3).

7 · Training: Preserve Known, Improve Novel



Known accuracy tracks the baseline while Novel accuracy rises.
 Surprise S increases as knownness k stays low - the behavior expected from selective complementary updates.

8 · Mechanism, Geometry, and Cost



Evidence slots localize reusable morphology;
 UMAP neighborhoods become more compact;
 Cost: 92.10 M -> 99.68 M parameters, +8% training, +6% inference.

A unified principle:
 discover what the dominant representation leaves unexplained.

UNIFYING FAILURE MODE — Rare tails, subtle abnormalities, and unseen siblings all hide decisive morphology behind a strong known pattern.
DESIGN CONSEQUENCE — SCAN estimates that dominant anchor, isolates high-energy residuals, and admits only orthogonal complementary evidence, so novel clusters can form without erasing known structure.
BOUNDARY OF THE CLAIM — Cross-modal, cross-host gains support a plug-in interpretation; remaining regressions show SCAN can recover a cue only when the backbone still encodes it.